Track 4: Renewable Energy and Green Technology

This track addresses the energy needs of farms and agricultural enterprises through renewable and sustainable technologies. Participants should consider energy availability, cost, storage, and the environmental benefits of their proposals, and should state their assumptions about energy supply and demand.

Problem Title 1: Renewable Energy-Based Smart Agricultural Drying System

A. Problem Description

Drying is energy-intensive and critical for quality and safe storage. Open-air drying depends on the weather and exposes produce to dust, rain, and pests, while electric or fuel-based dryers raise operating costs. Many farmers would benefit from a dryer that uses solar or other renewable heat.

Different products require different drying conditions and moisture targets. For example, spices, coffee, and grains each have preferred temperature ranges and final moisture levels, so a useful system must allow product-specific settings.

B. Challenges

1. Variability of solar energy with weather and time of day, and the need for backup or heat storage.
2. Meeting product-specific temperature, humidity, and moisture targets.
3. Ensuring uniform drying and airflow across the batch.
4. Balancing capital cost against energy savings for small farmers.
5. Monitoring moisture content reliably and protecting quality from overheating.

Problem Title 2: Solar-Powered Smart Irrigation and Water Pumping System

A. Problem Description

Irrigation pumps often depend on grid electricity, which may be unreliable or available only at fixed hours, or on diesel, which is costly. Solar pumping can reduce running costs, but pumps may run when water is not needed, or fail to deliver enough water on cloudy days.

Pairing solar pumping with intelligent control can match water delivery to crop needs and to the available sunlight and water source.

B. Challenges

1. Matching pump capacity to variable solar power and to head and flow requirements.
2. Managing intermittent energy supply, with or without battery storage.
3. Protecting pumps against dry running and low water levels.
4. Cost and maintenance of panels, controllers, and pumps for small farms.
5. Integrating soil or crop information for efficient scheduling.

Problem Title 3: Renewable Energy-Based Agricultural Cold Storage

A. Problem Description

Fruits, vegetables, and flowers deteriorate quickly after harvest, and many farms lack access to affordable cold storage, particularly in areas with unreliable electricity. Conventional refrigeration consumes considerable energy and can be uneconomical for small producers.

A cold store powered by solar or other renewable energy, with good insulation and thermal storage, can extend the shelf life of produce while limiting operating costs.

B. Challenges

1. Meeting cooling demand with variable renewable supply and limited battery capacity.
2. Reducing heat loss through insulation and efficient design.
3. Maintaining both temperature and humidity within suitable ranges for different produce.
4. Cost of equipment relative to the value of stored produce.
5. Reliable monitoring and alerting during power fluctuations.

Problem Title 4: Solar-Powered Agricultural Waste-to-Energy System

A. Problem Description

Crop residues, husks, and animal and organic waste are often burned in the open or left to decompose, causing air pollution and wasting a potential energy resource. Properly managed, such waste can produce biogas, fuel, or heat and leave useful by-products such as digestate or compost.

Solar power can supply monitoring and control functions, helping operate such systems reliably in rural settings.

B. Challenges

1. Variability in feedstock type, quantity, and moisture content.
2. Maintaining suitable conditions such as temperature, pH, and gas pressure in the process.
3. Safe handling and storage of biogas or other fuels.
4. Size, cost, and maintenance of equipment for small farms.
5. Handling residues and by-products responsibly.

Problem Title 5: Renewable Energy-Powered Agricultural Processing System

A. Problem Description

Small processing units rely on grid power or diesel generators, both of which are costly and may be unreliable. Motors starting under load create high power peaks, and interruptions can disrupt work and damage equipment.

Renewable power, supported by suitable energy management, can reduce operating cost and improve reliability for local processing.

B. Challenges

1. Handling high starting currents and variable loads with solar or other limited sources.
2. Sizing generation and storage appropriately for expected workloads.
3. Selecting efficient motors and drives for the chosen process.
4. Ensuring safety, dust control, and reliability in rural environments.
5. Demonstrating cost-effectiveness for small enterprises.